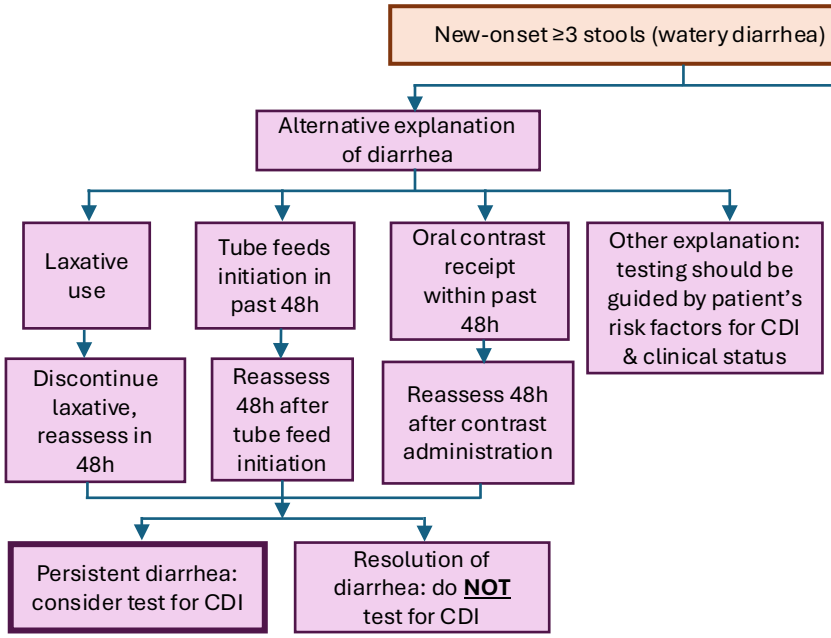
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Guideline for the Treatment of *Clostridioides difficile* in Adult Patients

Patients can be infected or colonized with *C. difficile* (*C. diff*). Available assays cannot differentiate between *C. diff* infection or colonization. For FY25, 34% of patients identified with *C. diff* infection did not meet criteria for diagnostic testing. Laxative use and <3 documented diarrheal stools were the most common causes for why *C. diff* tests were ordered inappropriately.

Refer to the C. difficile Treatment Guideline for treatment and testing guidance (see algorithm below).

Indications for *C. diff* Testing



Additional indications for *C. diff* testing:
- ✓ Colostomy output above baseline in 24h without alternate explanation
- ✓ Leukocytosis and ileus suspected to be caused by *C. diff*
- ✓ Acute abdomen with bowel thickening AND diarrhea or ileus
- ✓ Toxic megacolon or evidence of pseudomembranes

***C. diff* testing is NOT recommended for:**
- X Formed stool (patient is asymptomatic)
- X Diarrhea within 48h of laxative use, initiation of tube feeds, or oral contrast
- X Mild post-infectious IBS
- X As a test of cure
- X Currently on therapy for *C. difficile*
- X Within a 14-day period of a positive test
- X Within a 7-day period of a negative test
- X Children <1 year of age

Isolation precautions: Patients with a positive *C. diff* PCR and antigen test require Contact Precautions - Diarrheal for duration of admission.

C. diff Testing Methods:

- **Gastrointestinal Panel PCR plus *C. difficile* by PCR and Toxin bundle:** Recommended for pathogen identification of community-onset diarrhea on hospital days 1-3. Infectious Disease (ID) approval required if ordering on or after hospital day 4, given that the pathogens on this panel are unlikely causes of hospital onset diarrhea, even for immunocompromised patients.
- ***C. difficile by* PCR and Toxin:** Recommended for work up of hospital onset diarrhea (on or after hospital day 4), as this is the principle infectious etiology of hospital onset diarrhea.

Built-in MiChart Clinical Decision Support Tools		Recommended Action
***C. diff* Testing Warning OPA**	Alerts when a *C. diff* test is ordered for a patient admitted >72h AND has had one of the following in the previous 48h: initiation of new tube feedings, oral contrast, or laxatives.	Unless there is high clinical concern for *C. diff* infection (profuse watery diarrhea or acute ileus with abdominal pain and fevers), it is recommended to wait 48h after last administration before assessing for ongoing diarrhea and determining the need for *C. diff* testing.
***C. diff* Stool Documentation OPA**	Alerts when a *C. diff* test is ordered for a patient admitted >72h who has not met the minimum stool production frequency, quality, or volume based on documentation in the chart.	Confirm patient's stool frequency, quality, and volume. Discontinue *C. diff* test if stool documentation is accurate and there is not an alternative reason for testing.
Gastrointestinal Panel PCR Restriction	Any GI PCR panel that is ordered on or after day 4 of admission requires approval by the ID consult service.	If approved for GI PCR, enter the name of the ID physician consulted in the order.
***C. diff* by PCR and Toxin Re-testing Restriction**	Following a positive *C. diff* PCR result, MiChart will not allow ordering an additional test for 14 days.	Clinical decision making should consider the original test result and patient's signs and symptoms for acute infection.
***C. diff* Auto Discontinue Orders**	Gastrointestinal Panel PCR, C. diff by PCR and Toxin orders will be discontinued if stool is not collected within 24h of order placement.	

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Test Interpretation Guidance for Patients with Compatible Clinical Presentation

PCR Result	Antigen Result	Toxin Result	Interpretation	Recommendation
Negative	Not applicable	Not applicable	No C. diff present	Treatment: Not recommended, the negative predictive value of this test for ruling-out C. diff-associated diarrhea approaches 99% **Contact Precautions-D: Not required**
Positive	Negative	Negative	Low level C. diff colonization	Treatment: Not recommended **Contact Precautions-D: Not required**
Positive	Positive	Negative	C. diff infection or colonization may be present	Treatment: Clinically correlate to determine if treatment is warranted. Consider treatment with severe, non-resolving, or otherwise unexplained diarrhea strongly suggestive of C. diff infection **Contact Precautions-D: Yes, required for rest of admission**
Positive	Positive	Positive	C. diff infection present	Treatment: Recommended **Contact Precautions-D: Yes, required for rest of admission**

NOTE: Hyperlinks from document available here

- C difficile Treatment Guidelines
- Contact Precautions - Diarrheal

Treatment of *Clostridioides difficile* colitis

For all patients:
- Discontinue/change antibiotics if possible.
- Avoid PPI/H2 blockers without an appropriate indication.
- Implement infection control measures

Clinical Setting	Initial Episode[1,2]	First Recurrence[1,2,4] ≤ 90 days from prior episode	Second Recurrence[1,4,6] ≤ 90 days from prior episode
Non-Fulminant Disease Consider Infectious Diseases and Surgery consultation in patients with severe disease (defined as WBC ≥15,000, absolute neutrophil count ≤500, or SCr ≥1.5 times premorbid level)	Preferred treatment: **Vancomycin** 125 mg PO QID x 10 days Solid organ transplant patients:[5] **Fidaxomicin** 200 mg PO BID x10 days[7,8] OR **Vancomycin** 125 mg PO QID x10 days True vancomycin allergy (not vancomycin infusion reaction): **Fidaxomicin** 200 mg PO BID x10 days[7,8] **Metronidazole** 500 mg PO TID x10-14 days can be considered (third line option) in non-severe CDI if above agents are unavailable in outpatient setting	Infectious Diseases consultation is recommended **Fidaxomicin** 200 mg PO BID x5 days then 200 mg PO every other day x20 days[7,8] OR **Vancomycin** 125 mg PO QID x10 days then taper[6] over 3 weeks	Infectious Diseases consultation is strongly recommended The following options may be considered in consultation with Infectious Diseases[6]: - **(Preferred if not used yet): Fidaxomicin** 200 mg PO BID x5 days then 200 mg PO every other day x20 days[7,8] - **Vancomycin** taper[6] - Fecal microbiota transplant (see footnote 6c) - Kefir staggered protocol -
Fulminant Disease (Hypotension or septic shock, ileus, bowel obstruction, toxic megacolon, bowel perforation or peritonitis)	Surgery and Infectious Diseases Consultation are strongly recommended **Vancomycin** 500 mg PO QID + **Metronidazole** 500 mg IV q8h[3] If ileus, bowel obstruction, or fecal diversion, add **Vancomycin** by enema q6h Duration: Minimum of 14 days of therapy, depending on clinical response.	Surgery and Infectious Diseases Consultation are strongly recommended Repeat primary therapy then taper **vancomycin** over 5-11 weeks[6] Alternative if vancomycin taper cannot be performed: **Fidaxomicin** 200 mg PO BID x5 days then 200 mg PO every other day x20 days[7,8]	Surgery and Infectious Diseases Consultation are strongly recommended If vancomycin taper was NOT used for the first recurrence: Repeat primary therapy then taper[6] **vancomycin** over 5-11 weeks. If vancomycin taper was used for the first recurrence, the following options may be considered in consultation with Infectious Diseases[6]: - Repeat Vancomycin taper - **Fidaxomicin** 200 mg PO BID x5 days then 200 mg PO every other day x20 days[7,8] - Fecal microbiota transplant (see footnote 6c) - Kefir staggered protocol
	See footnote 1 for considerations in treatment refractory CDI. See footnote 11 for diverting loop ileostomy protocol.		

Initial/Recurrent CDI Considerations

1. Failure is defined as no improvement or worsening symptoms after 48-96 hours of primary therapy. In failing patients, look for alternative explanations/diagnoses, continue C. *difficile* treatment doses until resolution, and consider infectious diseases and gastroenterology consultation. Consider surgery consultation in patients progressing to fulminant disease. Changes in therapy (most commonly, switch to fidaxomicin in patients failing vancomycin) may be considered but there is not data to support this practice. In patients with a lack of alternative etiologies who are not responsive to standard pharmacologic therapies by day 5, FMT with inpatient-administered Rebyota (see 6c below) may be considered in consultation with Infectious Diseases and Gastroenterology.
2. Randomized trials have all utilized 10-day durations of therapy. Extension of course to 14 days may be considered in patients who have not had symptom resolution by day 10.
3. Parenteral administration of metronidazole has poor intraluminal penetration and should not be used alone for treatment. Parenteral vancomycin has no significant luminal accumulation and should not be used for C. difficile treatment.
4. Consider the use of vancomycin prophylaxis in patients that had a first or greater recurrence of CDI or fulminant disease in the past 90 days and require antimicrobials for a different infection. Other patients (including first occurrence of CDI) may be considered candidates for prophylaxis on a case-by-case basis in consultation with Infectious Diseases. The dose of prophylactic vancomycin is 125 mg daily to BID. **Using fidaxomicin as prophylaxis is not recommended.**
5. The choice of fidaxomicin or vancomycin for initial episode in solid organ transplant patients should consider the degree of immunosuppression, severity of disease, and concomitant administration of antibiotics.
6. Alternative and/or adjunctive agents:
    a. Vancomycin tapers should begin after the treatment course is completed. In a study of patients with 1st recurrence of CDI, a taper (started after 125mg PO QID X 10 days) of 125mg PO once daily X 7 days, 125mg PO once every other day X 7 days, and 125mg PO once every third day X 7 days was found to be efficacious. Patients with multiply recurrent disease may need more prolonged tapers. Such an example is : 125 mg PO BID x7 days, then 125 mg PO daily x7 days, then 125 mg PO every other day x7 days, then 125 mg PO every 3 days x2-8 weeks. Patients on tapered doses of PO vancomycin should continue to be monitored for signs and symptoms of C. difficile disease.
    b. Kefir staggered protocol: **Vancomycin** 125 mg QID x2 weeks, 375 mg q72h x2 weeks, 250 mg q72h x2 weeks, and 125 mg q72h x2 weeks PLUS kefir (5-oz glass with each meal (at least 3 glasses per day)) for 15 weeks.
    c. Fecal microbiota transplantation (FMT). There are 2 FMT preparations available at Michigan Medicine, live-jslm (Rebyota) and live-brpk (Vowst). Rebyota is the only FMT preparation available on the inpatient formulary at Michigan Medicine. Both Rebyota and Vowst are FMT options in the outpatient setting. Rebyota is administered via a single rectal installation, while Vowst is administered as 4 capsules PO QD X 3 days. Biobank derived FMT (OpenBiome) is no longer available at Michigan Medicine. FMT may be considered in the following scenarios.
        i. Recurrent CDI. FMT should be administered in the outpatient setting. The Vowst product is preferred. For outpatient use, patients should be referred to the infectious diseases or gastroenterology clinic. If possible, FMT should be deferred in patients requiring non-CDI antibiotic therapy to maximize therapeutic efficacy of microbiota replacement. In exceptional scenarios, Vowst may be available for inpatient administration via the non-formulary approval process.
        ii. CDI failing current therapy. See note 1 above. Infectious Diseases and Gastroenterology consultation are required. The Rebyota product is preferred.
        iii. Fulminant CDI. Infectious Diseases, Gastroenterology, and Surgery consultation is required. See Appendix for further details. Patients with bowel perforation are not candidates for FMT.
    d. The role of probiotics in prevention and treatment of *C. difficile* colitis is unclear, and their use is not currently recommended for inpatients. Avoid the use of probiotics in immunocompromised patients (transplant recipients, unintact gut mucosa, neutropenic patients, HIV/AIDS patients, etc) and patients with severe C. *difficile* colitis.
    e. Cholestyramine binds PO vancomycin and may decrease its efficacy. Avoid concomitant use.

Outpatient Treatment Considerations

7. A patient assistance program is available for fidaxomicin (https://www.merckconnect.com/dificid/patient-assistance.html?hcpUser=yes). Note that generic products are also now available.
8. For fidaxomicin use, outpatient coverage should be verified at time of treatment initiation by ordering 'Consult to Transitions of Care Pharmacy (TOC) Adult
9. For VOWST, financial assistance may be available for eligible patients. Further assistance can be accessed through either enrolling a patient in the VOWST Voyage Prescriber Portal or filling out the enrollment form (https://www.vowsthcp.com/patient-enrollment-support).

Fulminant Disease/Surgical Considerations

10. Intracolonic vancomycin 500 mg in 500 mL of normal saline every 6 hours given as retention enema using the following procedure: 18-inch Foley catheter with a 30-ml balloon inserted into rectum, balloon inflated, vancomycin instilled, catheter clamped for 60 minutes, deflate and remove. In patients who do not have the entire colon in place (i.e., a colorectal stump due to Hartman's procedure), a smaller volume of enema (100 mL) is acceptable
11. Postoperative diverting loop ileostomy regimen consists of antegrade vancomycin flushes (500 mg in 500 mL of Lactated Ringers; q8 hours for a duration of 10 days) via a 24 French Malecot catheter in the efferent limb of the ileostomy and intravenous (IV) metronidazole (500 mg q8 hours) for 10 days. See Reference Neal MD, et al. Ann Surg 2011;254:423-7.

APPENDIX: Inpatient Fecal Microbiota Transplantation for the Treatment of *Fulminant C difficile* Infection

Notes: The protocol was adapted from Ianiro G et al., 2018; PMID: 29851107. Reviewed and drafted by: Patricia Bloom, John Kao, Krishna Rao, and Danielle Kim Turgeon

Eligibility:

- Inpatient adult (≥18 years) with complicated CDI (CDI + hypotension attributed to CDI, ileus, megacolon, and/or peritonitis without perforation)
- Approval from both inpatient ID and GI consult services (or the GI attending if admitted to a GI service)

Exclusion Criteria:

- Current bowel perforation or perforation risk too high for colonoscopy OR enema as determined by GI/surgery (if enema delivery is deemed safe, then could proceed with that route)

Preparation:

- Anti-CDI therapy: this is per the discretion of the ID and GI services. Some patients can safely hold vancomycin prior to FMT while others cannot and should continue it until the FMT occurs.
- Bowel preparation: This is per the discretion of the endoscopist as some patients can safely tolerate a slow prep with at most 2 liters of PEG (either Golytely or Nulytely, per endoscopist preference) before the first procedure. If one goal is to assess for pseudomembranes then avoiding a prep is preferred.
- Restricted to a clear liquid diet
- Unlike for non-fulminant CDI, for fulminant CDI loperamide and other anti-motility agents should not be used for the procedure

Initial procedure description and instructions:

- Use at least 50 grams of Rebyota for each sample, though if safe can infuse the full 150 cc of Rebyota. Pediatric colonoscope or upper endoscope encouraged due to perforation risk. CO_2 insufflation per endoscopist preference.
- Deliver Rebyota through the operative channel of the scope after reaching the most proximal point of the large bowel deemed safe using 50 mL syringes filled with the Rebyota infusion.
- Withdraw colonoscope. Position patient to avoid stool leakage according to infusion site (e.g. reverse Trendelenburg and on right side for infusion in cecum). Consider positioning the patient accordingly before withdrawing the scope so reduce stool leakage. Transfer back to floor after post-endoscopy monitoring period completed.

Antimicrobials:

- Whether oral vancomycin should be continued or stopped post procedure is patient-dependent and should be discussed with the ID and GI consult services. Vancomycin is continued post-FMT in several studies, but a consensus on indication and duration is not available. In general, vancomycin was continued for 4-5 days after FMT, although this may depend on response to FMT. One group of investigators based the decision on whether to continue vancomycin post-FMT on whether pseudomembranes were seen during FMT.

Follow-up and Subsequent procedure(s):

- Studies differ on whether a second FMT after the original is used, so the incremental benefit is not clear and should be patient-specific, per the discretion of the ID and GI services. If the patient is still experiencing severe or complicated CDI 3 days after the initial procedure, can perform another FMT with the same protocol as above. If pseudomembranes are present on this exam, consider repeating endoscopy and performing FMT every 3 days until disappearance of the pseudomembranes, as was done in the Ianiro et al. study.
- Upon completion of additional FMT(s), outpatient follow-up with ID and/or GI may be needed.

Last revised: 08/2026	**Antimicrobial Stewardship Subcommittee:** 08/2026	**P&T:** 09/2026

References	
	Citation
1	Clinical Practice Guideline by the Infectious Diseases Society of America (IDSA) and Society for Healthcare Epidemiology of America (SHEA): 2021 Focused Update Guidelines on Management of Clostridioides difficile Infection in Adults. Available at: https://www.idsociety.org/practice-guideline/clostridioides-difficile-2021-focused-update/
2	Bakken JS. Staggered and Tapered Antibiotic Withdrawal With Administration of Kefir for Recurrent Clostridium difficile Infection. Clin Infect Dis 2014;59:858–61.
3	Neal MD et al. Diverting Loop Ileostomy and Colonic Lavage: An Alternative to Total Abdominal Colectomy for the Treatment of Severe, Complicated Clostridium Difficile Associated Disease. Ann Surg 2011;254 (3), 423-7.
4	Ianiro G et al., 2018; PMID: 29851107

Revision History	
Date	**Revision**
01/2020	Added diverting loop ileostomy footnote
07/2021	Updated testing criteria and process
09/2021	Updated vancomycin infusion reaction terminology
07/2022	Updated initial treatment options
03/2023	Revised FMT recommendations
02/2025	Removed bezlotoxumab. Updated FMT recommendations.
10/2025	Updated diagnostic recommendations
06/2026	Updated diagnostic considerations for new microbiological approach
08/2026	Updated 1st recurrence treatment recommendations

The recommendations in this guide are meant to serve as treatment guidelines for use at Michigan Medicine facilities. If you are an individual experiencing a medical emergency, call 911 immediately. These guidelines should not replace a provider's professional medical advice based on clinical judgment, or be used in lieu of an Infectious Diseases consultation when necessary. As a result of ongoing research, practice guidelines may from time to time change. The authors of these guidelines have made all attempts to ensure the accuracy based on current information, however, due to ongoing research, users of these guidelines are strongly encouraged to confirm the information contained within them through an independent source.